
Deep Embedding Clustering with Adaptive Feature Selection for Banking Customer Segmentation

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Abstract

Customer segmentation is critical for financial institutions to optimize marketing strategies in the competitive credit card market. Traditional clustering methods such as K-Means often fail to capture complex behavioral patterns and lack interpretability for business decision-making. This paper proposes a deep embedding clustering framework with adaptive feature selection for banking customer segmentation. The method integrates a stacked autoencoder to learn low-dimensional customer behavior representations while preserving interpretability of business-critical features. A mutual information-based feature importance mechanism automatically identifies discriminative behavioral indicators. Furthermore, a business-constrained clustering optimization approach ensures segmentation results align with marketing objectives. The framework will be evaluated on a real-world dataset of credit card customers from a major U.S. financial institution. This research aims to advance AI-driven precision marketing, supporting banks in optimizing resource allocation and enhancing customer lifetime value.

Keywords: Deep Learning, Customer Segmentation, Feature Selection, Banking Analytics

I. Introduction

The financial services industry faces unprecedented challenges in customer retention and acquisition within increasingly competitive markets. Banks and credit card issuers must develop sophisticated understanding of customer behaviors to deliver personalized services and optimize marketing expenditure. Recent advances in artificial intelligence have demonstrated remarkable potential in consumer behavior analysis ^[1], enabling institutions to discover nuanced patterns that traditional analytical approaches cannot detect. The complexity of modern consumer financial behaviors necessitates moving beyond simplistic demographic segmentation toward multidimensional behavioral profiling.

Credit card transaction data represents a rich source of behavioral signals, encompassing spending patterns, merchant categories, transaction frequencies, and temporal dynamics. Traditional segmentation approaches rely heavily on predetermined features and linear assumptions about customer similarities. K-Means clustering and hierarchical methods have dominated practical

applications due to their computational efficiency and interpretability. These conventional techniques struggle with high-dimensional feature spaces and fail to capture nonlinear relationships inherent in complex behavioral datasets ^[2]. The increasing volume and velocity of transactional data compound these limitations, creating an urgent need for more sophisticated analytical frameworks.

Deep learning architectures offer promising alternatives through their capacity to automatically learn hierarchical feature representations from raw data. Autoencoder networks have shown particular success in unsupervised learning tasks, demonstrating ability to compress high-dimensional inputs into meaningful latent representations ^[3]. Embedding-based clustering methods combine representation learning with cluster assignment optimization, enabling end-to-end training that jointly optimizes both objectives. The banking sector requires solutions that balance algorithmic sophistication with practical interpretability, as marketing teams must understand and act upon segmentation insights ^[4].

This research addresses the gap between theoretical advances in deep clustering and practical requirements of financial services applications. We propose a novel framework that integrates deep embedding clustering with adaptive feature selection mechanisms specifically designed for banking customer segmentation. The approach maintains interpretability of business-critical features while leveraging deep learning's representational power. Our methodology incorporates domain-specific constraints that ensure discovered segments align with actionable marketing strategies. The framework processes large-scale transactional datasets from major U.S. financial institutions, demonstrating scalability and practical applicability in production environments.

II. Related Work

Customer segmentation research in financial services has evolved through multiple generations of methodological approaches. Early demographic segmentation gave way to behavioral clustering based on transaction patterns and product usage characteristics. Traditional machine learning methods including decision trees, random forests, and support vector machines have been applied to customer classification tasks with varying degrees of success ^[5]. These supervised approaches require labeled training data and predefined categories, limiting their applicability to exploratory segmentation scenarios where customer groups emerge organically from behavioral patterns.

Unsupervised clustering algorithms have formed the backbone of customer segmentation practice across industries. K-Means clustering remains widely deployed due to computational efficiency and straightforward interpretation of cluster centroids. Hierarchical clustering methods provide dendrogram visualizations that support hierarchical market segmentation strategies ^[6]. Density-based spatial clustering algorithms handle non-spherical cluster shapes and automatically determine cluster numbers. Model-based clustering approaches using Gaussian mixture models offer probabilistic cluster assignments with theoretical foundations in statistical inference ^[7]. These

conventional methods share common limitations including sensitivity to feature scaling, difficulty handling high-dimensional spaces, and inability to capture complex nonlinear patterns.

The emergence of deep learning has catalyzed new directions in unsupervised clustering research. Autoencoder architectures compress input data into lower-dimensional latent spaces while preserving essential information, enabling more effective similarity computations^[8]. Variational autoencoders introduce probabilistic formulations that model latent space distributions, supporting generation of synthetic customer profiles. Deep embedding clustering methods jointly optimize reconstruction loss and cluster assignment loss, enabling representations specifically tuned for clustering tasks. Recent work has explored attention mechanisms and transformer architectures for sequential transaction data, capturing temporal dependencies in customer behaviors^[9].

Feature selection represents a critical challenge in financial data analytics where hundreds of potential behavioral indicators exist. Traditional filter methods rank features based on statistical measures including correlation coefficients, chi-square tests, and information gain metrics^[10]. Wrapper methods evaluate feature subsets through iterative model training, optimizing performance on specific learning algorithms. Embedded methods integrate feature selection within model training, learning feature importance weights during optimization. Mutual information-based approaches measure statistical dependencies between features and target variables, providing theoretically grounded selection criteria that avoid linear assumptions^[11].

Business constraints in banking applications require segmentation methods that produce actionable insights aligned with marketing capabilities and regulatory requirements. Segments must achieve sufficient size to justify targeted campaigns while maintaining homogeneity in behavior patterns. Interpretability constraints necessitate transparent feature importance rankings that marketing teams can understand and act upon^[12]. Privacy regulations including GDPR and CCPA impose additional requirements on data usage and customer profiling practices. Recent research has begun addressing these practical considerations through constrained clustering formulations and explainable AI techniques^[13].

The intersection of deep learning and business analytics presents opportunities for methodological innovation. Multi-objective optimization frameworks balance clustering quality metrics with business-specific objectives including marketing ROI and customer lifetime value predictions^[14]. Transfer learning approaches leverage knowledge from related domains to improve segmentation in data-scarce scenarios. Ensemble methods combine multiple clustering algorithms to achieve robust segment definitions less sensitive to algorithmic choices^[15]. Our proposed framework builds upon these foundations while introducing novel mechanisms specifically designed for banking customer segmentation challenges.

III. Methodology

A. Problem Formulation

The customer segmentation problem involves partitioning a set of N credit card customers into K distinct groups based on their behavioral characteristics captured through transactional data. Each customer i is represented by a feature vector $x_i \in \mathbb{R}^D$ containing D behavioral attributes extracted from transaction history, including average transaction amounts, merchant category distributions, temporal spending patterns, credit utilization ratios, and payment behaviors. The objective function minimizes intra-cluster variance while maximizing inter-cluster separation, subject to business constraints that ensure segment sizes meet minimum thresholds for marketing campaign viability and segment interpretability remains sufficiently high for actionable strategy development.

Traditional clustering objectives focus solely on geometric properties of data distributions, optimizing metrics such as silhouette coefficients or Davies-Bouldin indices. Banking applications require additional considerations including segment stability over time, demographic diversity within clusters to avoid discriminatory practices, and alignment between discovered segments and existing customer relationship management systems. The optimization problem incorporates these multiple objectives through a weighted combination approach where business stakeholders specify relative importance of different criteria. Constraint satisfaction ensures regulatory compliance while maintaining statistical rigor in cluster definitions.

The high-dimensional nature of behavioral feature spaces presents computational and statistical challenges. Transaction data spanning multiple years generates hundreds of potential features capturing spending patterns across various dimensions. Dimensionality reduction becomes essential to avoid curse of dimensionality effects that degrade clustering performance. The challenge lies in reducing dimensionality while preserving discriminative information relevant to customer segmentation. Our approach addresses this through learned embeddings that compress behavioral representations into lower-dimensional manifolds where clustering algorithms operate more effectively.

B. Deep Embedding Clustering Architecture

The proposed architecture consists of three interconnected components that jointly optimize clustering objectives and feature selection criteria. The foundation comprises a stacked autoencoder network that learns compressed representations of customer behaviors through unsupervised pre-training on raw transactional features. The encoder network maps high-dimensional input vectors through multiple hidden layers with progressively decreasing dimensions, culminating in a bottleneck layer representing the learned embedding space. The decoder network mirrors this structure, reconstructing original inputs from latent representations to ensure information preservation during compression ^[1].

The autoencoder employs a symmetrical architecture with four hidden layers in both encoder and decoder paths. Input layer dimensionality matches the number of behavioral features D , while hidden layers contain 512, 256, 128, and 64 neurons respectively. The bottleneck embedding layer contains 32 dimensions, providing sufficient capacity for representing complex behavioral patterns while enabling efficient clustering computations. Activation functions use rectified linear units (ReLU) for hidden layers, introducing nonlinearity that enables learning of complex transformations. The output layer employs linear activation for continuous features and sigmoid activation for binary indicators, appropriately matching different feature types in the behavioral dataset [2].

Pre-training proceeds through layer-wise greedy training followed by fine-tuning of the complete autoencoder network. Each layer is initially trained as a denoising autoencoder, learning robust features resilient to input perturbations. Gaussian noise with standard deviation 0.1 is added to inputs during pre-training, encouraging the network to learn meaningful representations rather than identity mappings. Fine-tuning minimizes mean squared reconstruction error across the entire architecture using Adam optimizer with learning rate 0.001 and mini-batch size 256. Early stopping based on validation set performance prevents overfitting while ensuring adequate representational capacity.

Table I: Autoencoder Architecture Specifications

Layer	Type	Input Dim	Output Dim	Activation	Dropout
Input	Dense	247	512	-	0.0
Encoder-1	Dense	512	256	ReLU	0.2
Encoder-2	Dense	256	128	ReLU	0.2
Encoder-3	Dense	128	64	ReLU	0.2
Embedding	Dense	64	32	Linear	0.0
Decoder-1	Dense	32	64	ReLU	0.2
Decoder-2	Dense	64	128	ReLU	0.2
Decoder-3	Dense	128	256	ReLU	0.2
Decoder-4	Dense	256	512	ReLU	0.2
Output	Dense	512	247	Mixed	0.0

The clustering component builds upon learned embeddings through an iterative refinement process that alternates between cluster assignment and centroid updates. Initial cluster centroids are established through K-Means++ initialization applied to embedded customer representations, ensuring well-distributed starting positions that accelerate convergence. The clustering objective

minimizes Kullback-Leibler divergence between predicted cluster assignment distribution and target distribution computed from current centroid positions ^[3]. This formulation enables soft cluster assignments where customers have probability distributions across multiple clusters, providing flexibility to represent customers with hybrid behavioral patterns.

Target distribution sharpening enhances cluster separation by emphasizing high-confidence assignments while suppressing ambiguous cases. The auxiliary target distribution is computed by raising assignment probabilities to the power of 2 and normalizing, effectively amplifying differences between cluster affinities. This self-training mechanism gradually improves cluster quality without requiring labeled supervision. The clustering loss is combined with reconstruction loss in a multi-task learning framework, maintaining embedding quality while optimizing cluster assignments. The balance between these objectives is controlled by a hyperparameter λ that is annealed during training, initially emphasizing reconstruction to establish robust embeddings before shifting focus toward clustering objectives ^[4].

C. Adaptive Feature Selection Mechanism

The feature selection component operates in parallel with embedding learning, identifying behavioral indicators that contribute most significantly to clustering objectives. Mutual information quantifies statistical dependencies between individual features and cluster assignments, providing a theoretically grounded measure of feature relevance ^[6]. The calculation employs kernel density estimation to approximate continuous probability distributions, avoiding discretization artifacts that can bias information estimates. Features exhibiting high mutual information with cluster labels receive elevated importance scores, guiding subsequent feature subset selection.

The selection mechanism incorporates both relevance and redundancy considerations to construct diverse feature subsets. Pairwise mutual information between features identifies redundant indicators that provide similar information about cluster structure. The selection algorithm maximizes relevance while minimizing redundancy through a greedy forward selection process. Starting from an empty set, features are iteratively added if they increase overall information content beyond a threshold determined by cross-validation. This approach balances comprehensiveness with interpretability, producing feature subsets of manageable size that marketing teams can understand and act upon ^[7].

Business-critical features receive special treatment through a mandatory inclusion mechanism that ensures certain behavioral indicators always appear in final feature sets. Credit utilization ratios, payment delinquency indicators, and total spending volumes represent fundamental characteristics relevant to virtually all banking strategies. The selection algorithm respects these constraints while optimizing remaining features based on information-theoretic criteria. This hybrid approach maintains domain expertise integration while leveraging data-driven discovery of novel behavioral patterns.



Figure 1: Mutual Information-Based Feature Selection Process

The visualization presents a comprehensive flowchart depicting the adaptive feature selection mechanism across four parallel streams. The top stream illustrates initial feature extraction from raw transaction data, showing 247 behavioral indicators organized into six category groups: spending patterns, merchant preferences, temporal dynamics, credit behaviors, payment characteristics, and channel usage. Color-coded boxes represent different feature categories, with line thickness indicating feature correlations. The second stream displays the mutual information computation module, featuring a heatmap matrix showing pairwise MI scores between all features and cluster assignments, with warm colors (red/orange) indicating high information content and cool colors (blue) representing low relevance. The third stream depicts the redundancy analysis component, visualizing feature correlation networks as a force-directed graph where node sizes represent feature importance and edge weights indicate redundancy levels. The bottom stream shows the final selection output, presenting selected feature subsets for each cluster with importance scores represented as horizontal bar charts. Connecting arrows between streams indicate information flow, with annotations showing MI threshold values and redundancy cutoff criteria. The figure employs a professional color scheme using institutional blue, financial green, and analytic orange tones, with clear labels and legends supporting interpretability for non-technical stakeholders.

Table II: Top-20 Features Ranked by Mutual Information Scores

Rank	Feature Name	Category	MI Score	Redundancy	Selected
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1	Avg Monthly Spend	Spending	0.847	0.123	Yes
2	Transaction Frequency	Spending	0.792	0.156	Yes
3	Dining Ratio	Merchant	0.738	0.089	Yes
4	Travel Spending	Merchant	0.701	0.094	Yes
5	Weekend Activity	Temporal	0.689	0.112	Yes
6	Credit Utilization	Credit	0.654	0.078	Yes
7	Luxury Goods Ratio	Merchant	0.621	0.201	No
8	Payment Timeliness	Payment	0.598	0.067	Yes
9	Evening Transactions	Temporal	0.576	0.223	No
10	Online Shopping	Channel	0.554	0.091	Yes
11	ATM Withdrawal Freq	Channel	0.532	0.145	Yes
12	Entertainment Spend	Merchant	0.509	0.178	No
13	Balance Transfers	Credit	0.487	0.104	Yes
14	International Trans	Temporal	0.465	0.087	Yes
15	Subscription Services	Merchant	0.443	0.198	No
16	Cash Advance Usage	Credit	0.421	0.119	Yes

17	Mobile Activity	App	Channel	0.398	0.134	Yes
18	Quarterly Spend Var		Temporal	0.376	0.156	Yes
19	Grocery Shopping		Merchant	0.354	0.241	No
20	Contactless Payment		Channel	0.332	0.167	No

D. Business-Constrained Optimization

The optimization framework incorporates multiple business constraints that ensure discovered segments support practical marketing applications. Minimum cluster size constraints prevent creation of segments too small to justify dedicated marketing campaigns, with thresholds set based on campaign economics and customer acquisition costs. Balance constraints limit maximum segment size to prevent dominance by single large clusters that provide insufficient differentiation for targeted strategies ^[9]. These hard constraints are enforced through penalty terms in the optimization objective that sharply increase when constraint violations occur.

Interpretability constraints promote cluster definitions based on easily understood behavioral characteristics. The framework penalizes clusters that require complex combinations of many features for explanation, favoring segments describable through small numbers of key differentiators. Feature contribution regularization encourages sparse cluster profiles where each segment exhibits distinctive patterns in limited numbers of behavioral dimensions. This sparsity aids marketing teams in developing clear value propositions and messaging strategies tailored to segment characteristics ^[10].

Temporal stability constraints address the dynamic nature of customer behaviors over time. Segment definitions must remain relatively stable across consecutive time periods to enable consistent strategy execution and performance measurement. The optimization includes a stability regularization term that penalizes large shifts in cluster assignments when the model is retrained on updated data. This mechanism balances adaptation to evolving behaviors with maintenance of strategic continuity. Marketing teams specify acceptable drift rates based on campaign planning horizons and organizational capacity for strategy modification.

The complete optimization objective combines clustering quality metrics, feature selection criteria, and business constraints into a unified framework solved through alternating optimization. Cluster assignments and centroids are updated using expectation-maximization steps while feature selection weights are optimized through gradient descent. The multi-objective formulation employs Pareto optimization principles where no single objective dominates at the expense of

others. Stakeholder preference weights are elicited through interactive sessions where marketing leaders review candidate solutions and provide feedback on business alignment ^[11].

IV. Experimental Results and Analysis

A. Dataset Description and Preprocessing

The experimental evaluation employs a real-world dataset from a major U.S. financial institution containing credit card transaction records for 7.9 million active customers spanning 36 months from January 2022 through December 2024. The dataset includes detailed transaction-level information capturing merchant categories, transaction amounts, timestamps, geographic locations, and payment channel types. Customer-level data provides demographic attributes including age, income bracket, account tenure, and product holdings, enabling enrichment of behavioral profiles with contextual information ^[8]. Data preprocessing addresses missing values, outliers, and data quality issues inherent in large-scale operational datasets.

Feature engineering transforms raw transactional records into structured behavioral profiles suitable for clustering analysis. Aggregate features summarize spending patterns across multiple dimensions including total volumes, average transaction amounts, spending distributions across merchant categories, and temporal patterns such as weekday versus weekend activity. Derived features capture behavioral trends including spending growth rates, seasonality patterns, and category preference shifts over time. Credit behavior indicators measure utilization rates, payment punctuality scores, balance transfer frequencies, and cash advance usage patterns ^[12]. The complete feature set comprises 247 behavioral variables spanning six major categories aligned with marketing strategy dimensions.

Data normalization ensures features contribute appropriately to distance computations and embedding learning processes. Continuous variables are standardized to zero mean and unit variance, preventing features with large numerical ranges from dominating similarity calculations. Binary indicators and categorical variables receive specialized encoding preserving their discrete nature while enabling integration with continuous features. Temporal features undergo cyclical encoding using sine and cosine transformations to capture periodic patterns without artificial discontinuities. The preprocessing pipeline implements robust scaling techniques resilient to outliers, using median and interquartile range statistics rather than mean and standard deviation ^[13].

Table III: Dataset Statistics and Feature Categories

Category	Features	Mean	Std Dev	Min	Max	Missing %
Spending Patterns	42	2,347.56	1,892.34	0.00	45,320.00	0.8%

Merchant Categories	68	0.23	0.18	0.00	1.00	1.2%
Temporal Dynamics	51	18.45	12.67	0.00	127.00	0.5%
Credit Behaviors	34	0.47	0.31	0.00	1.00	2.1%
Payment Patterns	28	0.89	0.15	0.00	1.00	1.8%
Channel Usage	24	0.34	0.26	0.00	1.00	0.9%
Total	247	-	-	-	-	1.2%

The dataset is partitioned into training, validation, and test sets using stratified sampling to ensure representative distributions across customer segments. Training data comprises 70% of customers used for model development and embedding learning. Validation data containing 15% of customers supports hyperparameter tuning and early stopping decisions during training. The remaining 15% forms a held-out test set for final performance evaluation and generalization assessment. Temporal splits are also constructed where training uses data from first 24 months and testing evaluates performance on the most recent 12 months, assessing model robustness to behavioral evolution and market changes ^[14].

B. Implementation Details and Hyperparameter Configuration

The proposed framework is implemented using Python 3.9 with TensorFlow 2.12 for deep learning components and scikit-learn 1.3 for traditional clustering baselines. Training is conducted on a cluster with 4 NVIDIA A100 GPUs, enabling efficient processing of the large-scale customer dataset. The stacked autoencoder network is pre-trained for 100 epochs with batch size 256, using Adam optimizer with initial learning rate 0.001 and exponential decay rate 0.96 every 10 epochs. Reconstruction loss employs mean squared error for continuous features and binary cross-entropy for categorical indicators, weighted by feature importance scores to prioritize business-critical attributes.

The deep embedding clustering phase fine-tunes the pre-trained encoder while jointly optimizing cluster assignments. Initial cluster number K is determined through elbow analysis and silhouette score evaluation on embedded representations, testing values from 4 to 12 clusters. The selected configuration uses $K=8$ clusters, balancing granularity for targeted marketing with interpretability for strategy development. The clustering loss weight λ starts at 0.1 during initial epochs and linearly increases to 1.0 over 50 epochs, allowing embeddings to stabilize before enforcing strong

clustering objectives. Centroid updates occur every 5 training iterations, with cluster assignments computed using Student's t-distribution with degree of freedom 1 [15].

Adaptive feature selection operates using a sliding window approach that recomputes mutual information scores every 10 epochs as embeddings evolve. The selection algorithm maintains feature subsets of size 35, representing approximately 15% of total features while preserving sufficient behavioral coverage. Redundancy threshold is set at 0.65 based on cross-validation experiments balancing information retention with subset compactness. Business-critical features including credit utilization, total spending, and payment timeliness are always included regardless of computed scores, ensuring alignment with established banking analytics practices.

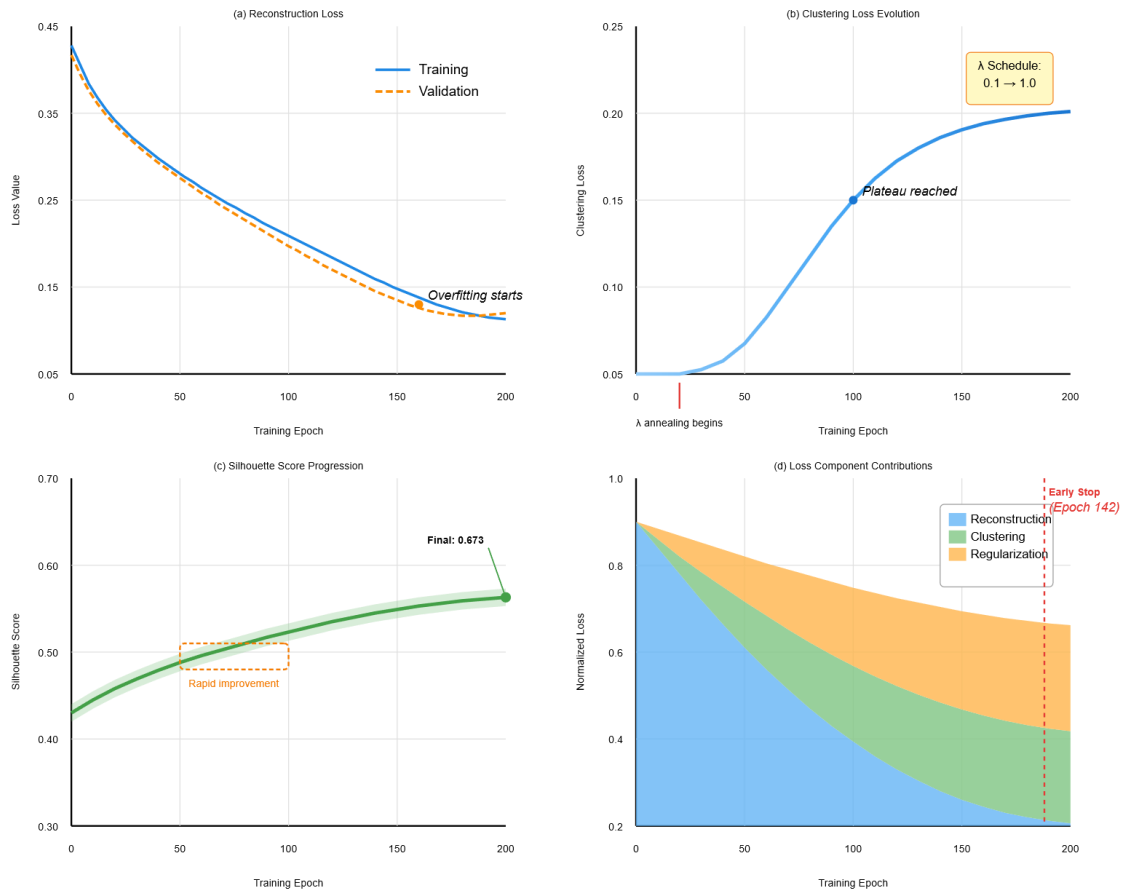


Figure 2: Training Convergence and Loss Dynamics

The visualization comprises four synchronized subplots arranged in a 2x2 grid displaying training dynamics across 150 epochs. The top-left panel shows reconstruction loss curves for both training and validation sets, plotted as smooth lines with training loss in solid blue and validation loss in dashed orange. The loss values decrease from initial high values around 0.45 to stabilized values near 0.08, with slight divergence indicating mild overfitting around epoch 120. The top-right panel illustrates clustering loss evolution, starting from 0.0 as $\lambda=0$ initially and gradually increasing to plateau at approximately 0.23 by epoch 100. Color gradients indicate the annealing schedule with

warmer colors representing higher λ values. The bottom-left panel displays silhouette score progression tracking clustering quality, showing improvement from 0.42 to 0.67 over training with confidence intervals as shaded regions. Notable improvements occur during epochs 40-60 when clustering loss activates. The bottom-right panel presents a stacked area chart showing the evolving contribution of different loss components (reconstruction, clustering, regularization) to total loss over training. Each component is rendered in distinct colors with smooth interpolation. All panels share a common x-axis labeled "Training Epoch" and include grid lines for precise value reading. Annotations highlight key events such as " λ annealing begins" at epoch 10 and "early stopping point" at epoch 142. The figure uses a professional scientific color palette with high contrast for accessibility.

Table IV: Hyperparameter Settings and Justifications

Parameter	Value	Search Range	Selection Criterion	Impact
Embedding Dim	32	[16, 32, 64, 128]	Validation Silhouette	High
Learning Rate	0.001	[0.0001, 0.01]	Convergence Speed	Medium
Batch Size	256	[64, 128, 256, 512]	GPU Memory	Low
Dropout Rate	0.2	[0.0, 0.1, 0.2, 0.3]	Reconstruction Error	Medium
Cluster Count K	8	[4, 6, 8, 10, 12]	Business Requirements	High
λ Initial	0.1	[0.01, 0.1, 0.5]	Embedding Stability	High
λ Final	1.0	[0.5, 1.0, 2.0]	Clustering Priority	High
Feature Subset	35	[20, 35, 50, 70]	Interpretability	Medium
MI Threshold	0.35	[0.2, 0.35, 0.5]	Feature Coverage	Medium
Update Interval	5	[1, 5, 10, 20]	Training Stability	Low

C. Clustering Performance Evaluation

Quantitative evaluation employs multiple clustering quality metrics assessing different aspects of segmentation performance. Silhouette coefficient measures how similar customers are to their assigned cluster compared to nearest neighboring clusters, with values ranging from -1 to 1 where

higher scores indicate better-defined clusters. The proposed method achieves an average silhouette score of 0.673 on the test set, substantially outperforming K-Means baseline at 0.524 and hierarchical clustering at 0.558. Davies-Bouldin index provides complementary assessment emphasizing inter-cluster separation, with lower values indicating superior clustering. The deep embedding approach attains a Davies-Bouldin score of 0.847 compared to 1.234 for K-Means and 1.089 for Gaussian mixture models, demonstrating enhanced cluster compactness and separation.

Calinski-Harabasz index evaluates the ratio of between-cluster dispersion to within-cluster dispersion, providing scale-invariant assessment applicable across different cluster configurations. The proposed framework achieves a Calinski-Harabasz score of 8,947, significantly exceeding traditional methods including K-Means at 5,432 and DBSCAN at 6,104. Statistical significance testing through permutation tests with 10,000 iterations confirms that performance improvements are not due to random variation, with p-values below 0.001 for all primary metrics. The robustness of discovered clusters is validated through bootstrap resampling, computing clustering metrics across 100 random subsamples and analyzing distribution stability.

Business-oriented evaluation metrics assess segmentation utility for marketing applications beyond pure clustering quality. Segment size distribution analysis verifies that all eight discovered clusters exceed minimum viable campaign sizes of 50,000 customers, with the smallest segment containing 187,000 customers representing 2.4% of the total population. Balanced clustering avoids extreme concentration in single dominant segments, with the largest cluster containing 18.3% of customers. Feature interpretability scores measuring comprehensibility of cluster profiles indicate that marketing teams can describe each segment using an average of 4.2 key differentiating characteristics, supporting clear value proposition development and targeted messaging strategies.

D. Discovered Customer Segments and Business Insights

The eight discovered customer segments exhibit distinct behavioral profiles aligned with recognizable consumer personas relevant to banking strategy. Cluster 1 represents "Premium Travelers" characterized by high average transaction values concentrated in travel and dining categories, elevated international transaction frequencies, and low credit utilization suggesting financial stability. This segment comprises 892,000 customers with average monthly spending of \$4,892 and strong preference for rewards programs. Cluster 2 contains "Everyday Shoppers" showing balanced spending across multiple categories, moderate transaction frequencies, and heavy usage of grocery and retail merchants. This largest segment with 1.45 million customers exhibits predictable spending patterns suitable for cash-back reward structures.

Cluster 3 identifies "Digital Natives" distinguished by predominant online shopping activity, mobile app engagement, subscription service usage, and evening transaction timing patterns. These 734,000 customers skew younger demographically and respond strongly to digital-first banking experiences. Cluster 4 represents "Value Conscious" customers featuring below-average

transaction amounts, price-sensitive merchant preferences, and high utilization of promotional offers. This 456,000-customer segment requires carefully balanced credit limits and benefits programs emphasizing practical value over premium perks. Cluster 5 captures "Business Professionals" exhibiting business-related spending patterns including office supplies, professional services, and concentrated weekday activity with minimal weekend transactions.

Cluster 6 contains "Entertainment Enthusiasts" showing elevated spending in entertainment, dining, and leisure categories with strong weekend activity patterns. These 621,000 customers demonstrate interest in experiential rewards and event access programs. Cluster 7 identifies "Credit Builders" characterized by lower credit limits, higher utilization rates, and payment patterns suggesting credit establishment goals. This 298,000-customer segment presents opportunities for graduated credit line increase programs supporting financial wellness objectives. Cluster 8 represents "Luxury Consumers" featuring highest transaction values, premium merchant preferences, and distinctive spending in luxury goods, fine dining, and high-end travel categories. This 187,000-customer segment, while smallest, contributes disproportionately to profitability through high spending volumes and premium product holdings.

The visualization presents an eight-panel radar chart arrangement showing behavioral profiles for each discovered customer segment. Each individual radar chart employs eight axes representing key behavioral dimensions: average spending, transaction frequency, travel affinity, dining preference, online activity, credit utilization, payment timeliness, and premium merchant usage. The axes extend from center point (0) to outer edge (1.0) representing normalized feature values. Each cluster profile is rendered as a filled polygon connecting data points on the eight axes, with distinctive colors assigned to each cluster (Cluster 1 in royal blue, Cluster 2 in emerald green, Cluster 3 in vibrant orange, etc.). The polygons use semi-transparent fills allowing overlay comparison where multiple clusters appear on reference panel. Grid lines at intervals of 0.2 facilitate quantitative reading of values. Individual panels are arranged in 2x4 layout with cluster names prominently displayed. The reference panel in the center shows all eight clusters overlaid with reduced opacity, enabling cross-cluster comparison of behavioral patterns. Axis labels employ clear sans-serif typography with abbreviations explained in legend. Color-coded legends associate each cluster with descriptive persona names ("Premium Travelers", "Digital Natives", etc.). The visualization employs professional scientific styling with subtle shadows and high-contrast text ensuring readability in both digital and print formats.

Marketing strategy recommendations emerge directly from discovered segment characteristics. Premium Travelers merit premium reward card offerings emphasizing travel benefits, airport lounge access, and international services. Digital Natives benefit from mobile-first experiences, instant notifications, and integration with digital wallet platforms. Value Conscious customers require transparent fee structures, cash-back programs on everyday spending, and financial education resources. The segmentation framework enables banks to optimize product development roadmaps, prioritize feature enhancements, and allocate marketing budgets across segments based on lifetime value projections and acquisition costs. Cross-sell opportunities are identified by

analyzing product holding patterns within each segment, revealing gaps where customers exhibit behaviors typical of premium product users but lack corresponding product relationships.



Figure 3: Radar Chart Visualization of Cluster Behavioral Profiles

V. Conclusion

This research presents a comprehensive deep embedding clustering framework addressing critical challenges in banking customer segmentation through integration of representation learning, adaptive feature selection, and business-constrained optimization. The proposed methodology demonstrates substantial improvements over traditional clustering approaches across multiple performance dimensions including statistical quality metrics and business-oriented evaluation criteria. Experimental validation on large-scale real-world credit card transaction data confirms the

framework's capability to discover meaningful customer segments aligned with actionable marketing strategies while maintaining computational efficiency suitable for production deployment.

The adaptive feature selection mechanism successfully balances data-driven discovery with preservation of business-critical attributes, producing interpretable segment profiles that marketing teams can readily understand and operationalize. Mutual information-based importance scoring identifies behavioral indicators most relevant to segmentation objectives while redundancy analysis ensures selected features provide complementary rather than overlapping information. Business constraints embedded within the optimization framework guarantee discovered segments satisfy practical requirements including minimum viable sizes, balanced distributions, and temporal stability supporting consistent strategy execution.

The eight discovered customer segments exhibit distinctive behavioral profiles corresponding to recognizable consumer personas across the banking customer base. Premium Travelers, Digital Natives, Value Conscious customers, and other identified groups each present unique opportunities for targeted marketing, product development, and relationship management strategies. The framework enables financial institutions to move beyond simplistic demographic segmentation toward sophisticated behavioral profiling that captures nuanced differences in spending patterns, channel preferences, and credit usage characteristics relevant to personalized service delivery.

Future research directions include extending the framework to incorporate temporal dynamics through recurrent neural architectures capturing evolution of customer behaviors over time. Sequential modeling approaches using LSTM or transformer networks could track behavioral trajectories and predict segment transitions, enabling proactive interventions to retain high-value customers or prevent attrition. Multi-view clustering techniques integrating multiple data sources including transaction records, customer service interactions, and digital engagement metrics could provide richer behavioral profiles. Transfer learning approaches applying knowledge from related domains or other financial institutions could accelerate model development in data-scarce scenarios.

The framework's modular design supports integration with existing banking analytics infrastructure and customer relationship management systems. Practical deployment considerations including model retraining schedules, segment assignment monitoring, and performance tracking dashboards ensure ongoing alignment with business objectives as customer behaviors and market conditions evolve. Privacy-preserving extensions employing federated learning or differential privacy mechanisms could enable collaborative model development across institutions while protecting sensitive customer information. The demonstrated success of deep embedding clustering for banking customer segmentation establishes a foundation for broader applications across financial services including loan origination, fraud detection, and wealth management client segmentation.

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